

Google OAuth Setup — Step by Step

How the Google Cloud project for SparkClone was created · Account: kbalaji9738@gmail.com · August 11, 2026 · Updated August 12, 2026 (APIs)

This guide records exactly how SparkClone's Google integration was configured, so you can repeat it for any future project (or another Google account). Everything below is free — no billing account or credit card is required for these APIs at personal usage levels.

What you end up with: an OAuth Client ID and Client Secret — a pair identifying your app to Google so it can show the "Allow this app to access your Gmail/Drive?" consent screen — plus an API key for the YouTube Data API. The OAuth pair is not the same as an API key: no password or key of yours is ever given to the app; Google issues limited-power tokens after you approve.

Part 1 — Create the project

STEP 1 — OPEN THE PROJECT CREATION PAGE

Go to <https://console.cloud.google.com/projectcreate> and make sure the account menu (top-right avatar) shows the Google account you want the agent to access — here, kbalaji9738@gmail.com. If another account is active, click the avatar and switch accounts first.

STEP 2 — NAME AND CREATE IT

Project name: **sparkclone** · Parent resource: **No organization** (default for personal Gmail accounts). Click **Create**, wait a few seconds, then pick the new project from the project dropdown (top-left). Result: project ID **sparkclone-505213**.

Personal accounts have a default quota of ~25 projects. This account had 1 slot remaining, which this project used — delete unused projects at IAM & Admin → Manage Resources if you ever need another.

Part 2 — Enable the APIs

All of these are enabled the same way: Menu → APIs & Services → Library, search the API by name, open it, click **Enable**. Six APIs are enabled for this project:

STEP 3 — GMAIL API

Search "Gmail API", click Enable. Powers inbox reading and (approval-gated) sending.
(console.cloud.google.com/apis/library/gmail.googleapis.com)

STEP 4 — GOOGLE DRIVE API

Search "Google Drive API", click Enable. Powers file listing/reading and Docs export.
(console.cloud.google.com/apis/library/drive.googleapis.com)

STEP 5 — YOUTUBE DATA API V3

Search "YouTube Data API v3", click Enable. Powers the agent's YouTube tools — video metadata, channel upload feeds, and transcript lookups. Unlike Gmail/Drive it is used with a plain **API key** (Step 11), not OAuth — it only reads public YouTube data, so no consent screen is involved.
(console.cloud.google.com/apis/library/youtube.googleapis.com)

STEP 6 — GOOGLE DOCS, CALENDAR AND SHEETS APIS

Enabled the same way, one after another: **Google Docs API** (docs.googleapis.com), **Google Calendar API** (calendar-json.googleapis.com) and **Google Sheets API** (sheets.googleapis.com).

Docs, Calendar and Sheets are enabled ahead of need: the SparkClone code currently requests only Gmail + Drive OAuth scopes, so these three cost nothing and sit idle until tools use them. When such a tool is added, extend the scope list in the backend and re-consent — no console work will be needed.

Part 3 — Configure the OAuth consent screen

STEP 7 — START THE WIZARD

Menu → APIs & Services → OAuth consent screen (Google now calls this "Google Auth Platform"). Click **Get started**. Fill the four sections:

App Information	App name SparkClone , support email kbalaji9738@gmail.com
Audience	External (personal Gmail accounts can't use Internal)
Contact Information	kbalaji9738@gmail.com
Finish	Tick "I agree to the Google API Services: User Data Policy", Continue, Create

STEP 8 — ADD YOURSELF AS A TEST USER

Go to the **Audience** page (left sidebar). Under Test users click **+ Add users**, enter kbalaji9738@gmail.com, and Save. The app stays in **Testing** publishing status — only listed test users can sign in, and no Google verification review is needed.

The one Testing-mode quirk: for External apps in Testing, Google expires refresh tokens after about 7 days. Practically: you'll click "Sign in with Google" in SparkClone again roughly once a week. Acceptable for a personal self-hosted tool; the alternative (publishing to Production) triggers a Google review because Gmail scopes are "restricted", which isn't worth it here.

Part 4 — Create the OAuth client

STEP 9 — CREATE THE CLIENT ID

Sidebar → **Clients** → **+ Create client**:

- Application type: **Web application**
- Name: **sparkclone-web**
- Authorized redirect URIs → **+ Add URI**: `http://localhost:8000/auth/google/callback` (this is the FastAPI route Google sends the user back to after consent — it must match the backend exactly, including port and path)
- Authorized JavaScript origins: left empty (not needed for a server-side flow)

Click **Create**.

STEP 10 — COPY THE CREDENTIALS IMMEDIATELY

A dialog shows the Client ID and Client Secret. The secret is shown only once — copy both (or Download JSON) before clicking OK. If you lose the secret, open the client on the Clients page and use Reset secret.

Client ID	145271900279-gq1dvut9hkc21kkjk873v68u766jg1ld.apps.googleusercontent.com
Client secret	GOCSPX-***** (stored in the project's .env; not printed here since PDFs get shared)

Part 5 — Create the YouTube API key

STEP 11 — API KEY FOR THE YOUTUBE DATA API

Sidebar → **Credentials** (under APIs & Services) → **+ Create credentials** → **API key**. Copy the AIza... key from the dialog. Then click the new key to edit it and restrict it: under **API restrictions** choose "Restrict key" and tick only **YouTube Data API v3**, so a leaked key can't touch anything else. Application restrictions stay on "None" for a server-side personal tool.

The free YouTube Data API quota is 10,000 units/day — a video-metadata lookup costs 1 unit, a search 100. SparkClone's per-video and channel-feed tools sit far below the ceiling at personal usage.

Part 6 — Wire it into SparkClone

STEP 12 — PUT THE VALUES IN .ENV

```
G00GLE_CLIENT_ID=145271900279-gq1dvut9hkc21kkjk873v68u766jg1ld.apps.googleusercontent...
G00GLE_CLIENT_SECRET=<the GOCSPX-... secret>
OAUTH_REDIRECT_URL=http://localhost:8000/auth/google/callback
YOUTUBE_API_KEY=<the AIza... key from Step 11>
# random values (generated with: openssl rand -hex 24)
SPARK_SECRET_KEY=<48 hex chars> # signs session cookies + encrypts stored refresh tokens
SPARK_API_TOKEN=<48 hex chars> # bearer-token fallback for scripts/API clients
```

STEP 13 — THE IMAP/SMTP BLOCK: LEAVE IT EMPTY

The .env still contains IMAP_HOST / SMTP_HOST / ... fields. These power the old email path (app-password based). With Google OAuth, the Gmail API replaces them, so they stay blank as an optional fallback. NOTIFY_EMAIL is kept filled so notifications know their destination. Until SMTP is configured or Gmail sending is built, the notify tool logs to the server console instead of emailing.

Result

Google Cloud project	sparkclone (sparkclone-505213) — Created
Gmail API + Google Drive API	Enabled (OAuth flow)
YouTube Data API v3	Enabled (API key, restricted to this API)
Docs + Calendar + Sheets APIs	Enabled (ahead of need; no scopes requested yet)
OAuth consent screen	"SparkClone" (External, Testing) — Configured
Test user	kbalaji9738@gmail.com — Added
OAuth client	sparkclone-web + localhost redirect URI — Created
YouTube API key	Restricted to YouTube Data API v3 — Created
Credentials saved to project .env	Done
Cost	\$0 (free) — no billing account needed